

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence. Where no observation exists, emem signs an absence and its reason, not a zero.

A compact citation keeps the observation traceable across agents and time. Design goal: encode in orbit, decode with AI.

Jaya Kumari · Avijeeet Singh · Vortex AI · avijeeet@vortex.ai · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026

Agent A cites a satellite observation. What can agent B check without trusting A or A's model?
emem makes satellite observations, raw or derived, addressable by place, band and time. Each is a record named by the BLAKE3 hash of its bytes and signed in a batch. The exact evidence A cites survives a change of model and a handoff; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name or signature. Trace token-record bytes-cell-pixel-receipt-log-key



From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · bind · sign · source Policy: refuse if one fails.	Reason with cited evidence Carry the reference onward.

1 Does agent B act on a corrupted handoff?

The same corruptions, five handoff conditions.

Agent A cites a satellite reading. A relay or a faulty signer changes the evidence. Agent B acts on it, declines it or acts on the genuine value.

Agent A cites	handoff	relay	Agent B	B did not act on corrupted evidence agents, pooled Claude	declined	acted on genuine value
 Agent A reads NDVI 0.4709 and cites it Sentinel-2A L2A, 25 Sep 2026 outlined: the 10 m pixel the record names RGB from recorded grids	prose The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha...	relay or faulty signer one change per trial, shared corruption families: value · unit · cell · time · band · source · derivation · pixel · stale record · signature / id	B reads the text	22 / 276	22	0
	JSON { "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448...		B reads the fields	112 / 276	106	6
	retrieved text (BM25) Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ...		B retrieves a passage	56 / 276	56	0
	opaque id record ref obs-7d1c44		B fetches the sender's record	134 / 288	88	46
	emem reference + verifier emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6...		B resolve re-hash place / time / band signature log recompute source re-read	300 / 300	264	36

Matched ablation, deterministic receiver, same 23 corruptions (4 Oct), accepted: JSON, no checks 23 → emem's checks minus hash, signature and log 8 → plus four consistency checks 3 → emem's verifier 0

Across the 23 shared items, agents with an emem reference did not act on 276 of 276 corrupted handoffs (240 declined, 36 acted on the genuine value); with an opaque id, 122 of 276; in prose, 22 of 276.

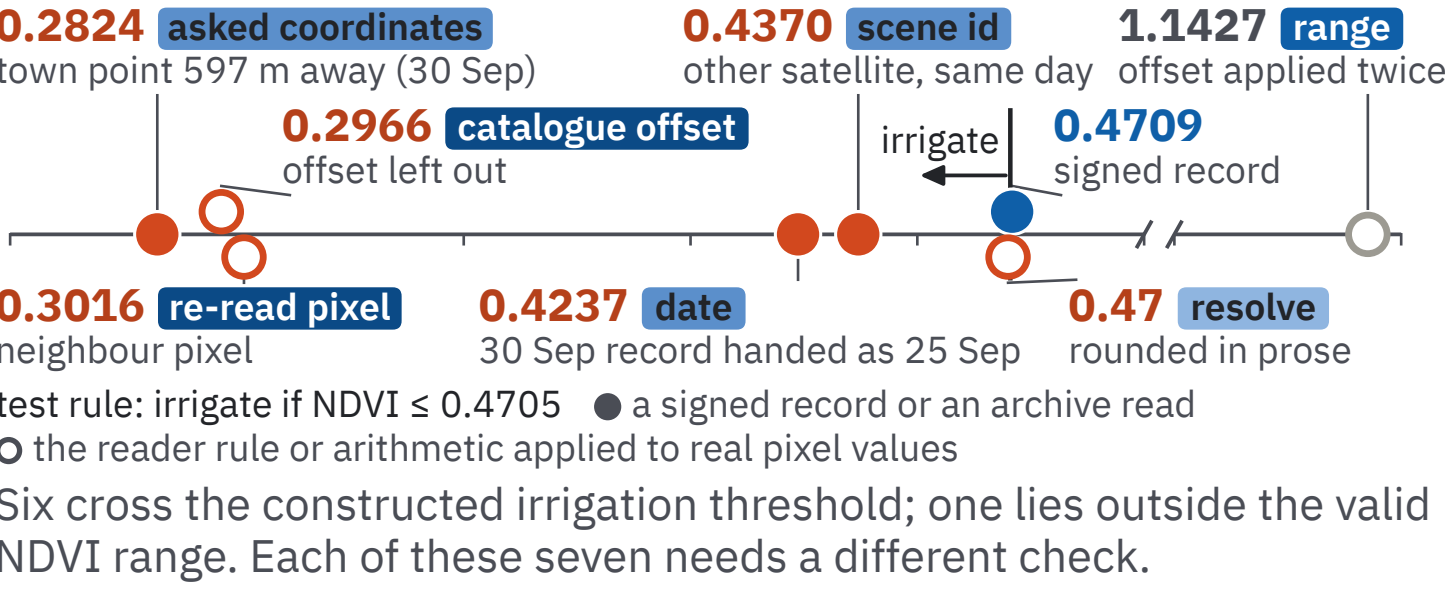
By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked, so the source is not re-read (SAT-042 harness, not flown).

Figure: Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5 pooled, full item set per condition. 2,794 scored trials (1 Oct 2026, two constructed tasks, Qwen included) span 25 corruption items, 2 controls and 1 exploratory item. The emem lane adds a verifier and an instruction, so the figure tests the workflow. With the ablation's matched checks, instructed Haiku agents acted on the same 8 (exploratory run, 4 Oct, one trial each). Qwen2.5-7B acted on 10 of 25 corruptions behind a fail-closed resolver (2 of 25 when told to check): gate the action outside the model.

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Why can a value change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta z = \Delta_{\text{env}} + \Delta_{\text{sensor}} + \Delta_{\text{geo}} + \Delta_{\text{encoder}} + \epsilon$$

Δz : change in readout. The terms name possible causes (environment, sensor, geolocation, encoder, residual); we have not measured their shares.

7 What does a checked reference establish?

The record, not physical truth

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token-record-source pixel

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG(row, col) = DN(b) & Ed25519(a) & leaf in STH & STHgeo.qa Prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... still trusted: BLAKE3 and the verifier's code

4-5 cell binding hashed cell = token cell = the lat/long the derivation used · the cell rule

6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code

8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · this STAC item carries no checksum

10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right

12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the signed tree head (STH) geo.qa co-signed · geo.qa is also run by Vortex AI

15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

CONTRIBUTION

Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem applies them to single EO observations, and its records name the source file and sampled point, so a receiver can re-read the pixel. That re-read caught a wrong pixel that hashing, binding, signatures, logs and recomputation had passed (panel 5).

12 Use emem. Carry the evidence forward.

ChatGPT · @emem plugin "Signed facts about real places" Enable emem; paste a token	Claude · Directory plugin / MCP "hand another agent a token it can check offline" Resolve from chat or terminal	Diffy · Marketplace plugin "Shared memory for AI agents. One address per fact, one signature you check." Workflow templates, including EUDR	GitHub MCP Registry · Server listing "Satellites for AI." Install MCP server	ClawHub · OpenClaw plugin listing "Shared, signed and content-addressed memory of the physical world..." Install the OpenClaw plugin	CONNECT Setup · EO tasks Reproduce
MORE CLIENTS Cursor · VS Code · Gemini CLI	PROTOCOLS MCP · A2A REST / OpenAPI	DEVELOPER Python · TypeScript · Docker LangGraph memory · GitHub source	FRAMEWORK EXAMPLES LangChain · LlamaIndex · AutoGen CrewAI · Mastra · Agno (see setup notes)	DISCOVERY MCP Registry · MuleSoft Exchange · Glama · Smithery Context7	